

Article

Lithium Conductivity and Meyer-Neldel Rule in Li_3PO_4 - Li_3VO_4 - Li_4GeO_4 Lithium Superionic Conductors

Sokseiha Muy, John C. Bachman, Hao-Hsun Chang, Livia Giordano, Filippo Maglia, Saskia Lupart, Peter Lamp, Wolfgang G. Zeier, and Yang Shao-Horn

Chem. Mater., **Just Accepted Manuscript** • DOI: 10.1021/acs.chemmater.8b01504 • Publication Date (Web): 26 Jul 2018

Downloaded from <http://pubs.acs.org> on July 27, 2018

Just Accepted

"Just Accepted" manuscripts have been peer-reviewed and accepted for publication. They are posted online prior to technical editing, formatting for publication and author proofing. The American Chemical Society provides "Just Accepted" as a service to the research community to expedite the dissemination of scientific material as soon as possible after acceptance. "Just Accepted" manuscripts appear in full in PDF format accompanied by an HTML abstract. "Just Accepted" manuscripts have been fully peer reviewed, but should not be considered the official version of record. They are citable by the Digital Object Identifier (DOI®). "Just Accepted" is an optional service offered to authors. Therefore, the "Just Accepted" Web site may not include all articles that will be published in the journal. After a manuscript is technically edited and formatted, it will be removed from the "Just Accepted" Web site and published as an ASAP article. Note that technical editing may introduce minor changes to the manuscript text and/or graphics which could affect content, and all legal disclaimers and ethical guidelines that apply to the journal pertain. ACS cannot be held responsible for errors or consequences arising from the use of information contained in these "Just Accepted" manuscripts.



ACS Publications

is published by the American Chemical Society, 1155 Sixteenth Street N.W., Washington, DC 20036

Published by American Chemical Society. Copyright © American Chemical Society. However, no copyright claim is made to original U.S. Government works, or works produced by employees of any Commonwealth realm Crown government in the course of their duties.

Lithium Conductivity and Meyer-Neldel Rule in $\text{Li}_3\text{PO}_4\text{-Li}_3\text{VO}_4\text{-Li}_4\text{GeO}_4$ Lithium Superionic Conductors

Sokseiha Muy,^{a,‡} John C. Bachman,^{b,c,†,*} Hao-Hsun Chang,^d Livia Giordano,^{b,e} Filippo Maglia,^f Saskia Lupart,^f Peter Lamp,^f Wolfgang G. Zeier,^g and Yang Shao-Horn^{a,b,d,*}

^aDepartment of Materials Science and Engineering, Massachusetts Institute of Technology, 77 Massachusetts Ave., Cambridge, MA, 02139, USA

^bDepartment of Mechanical Engineering, Massachusetts Institute of Technology, 77 Massachusetts Ave., Cambridge, MA, 02139, USA

^cDepartment of Mechanical Engineering, California State University, Los Angeles, 5151 State University Dr., Los Angeles, CA, 90032, USA

^dResearch Laboratory of Electronics, Massachusetts Institute of Technology, 77 Massachusetts Ave., Cambridge, MA, 02139, USA

^eDipartimento di Scienza dei Materiali, Università di Milano-Bicocca, 20125 Milano, Italy

^fResearch Battery Technology, BMW Group, Munich 80788, Germany

^gInstitute of Physical Chemistry, Justus-Liebig-University Giessen, Heinrich-Buff-Ring 17, D-35392 Giessen, Germany

ABSTRACT: The ionic conductivity and activation energy of lithium in the $\text{Li}_3\text{PO}_4\text{-Li}_3\text{VO}_4\text{-Li}_4\text{GeO}_4$ system was systematically investigated. The sharp decrease in activation energy upon Ge substitution in Li_3PO_4 and Li_3VO_4 was attributed to the reduction in the defect formation energy while the variation in activation energy upon increasing Ge content was rationalized in term of the inductive effect. We also found a correlation between the pre-exponential factors and the activation energies in agreement with the well-known Meyer-Neldel rule. The series of compound with and without partial lithium occupancy were shown to fall into two distinct lines. The slope of the line was found to be related to the inverse of the energy scale associated with phonons in the system, which agrees with the multi-excitation entropy theory. The intercept of the line was found to be related to the Gibbs free energy of defect formation. Compiled data of pre-exponential factor and activation energy for commonly studied lithium-ion conductors shows that this correlation is very general, implying an unfavorable trade-off between high pre-exponential factor and low activation energy needed to achieve high ionic conductivity. Understanding the circumstances under which this correlation can be violated might provide a new opportunity to further increase the ionic conductivity in lithium-ion conductors.

1. INTRODUCTION

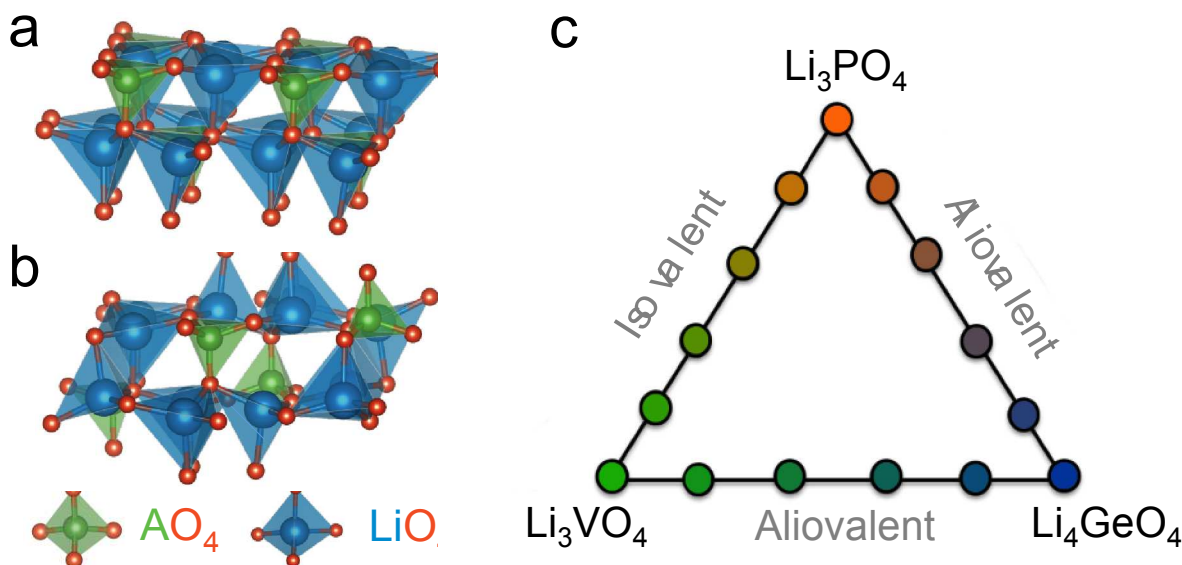


Figure 1. a) Crystal structure of the low-temperature structure of LISICON β - Li_3PO_4 (space group $\text{Pmn}2_1$).³² b) Crystal structure of the high-temperature structure of LISICON γ - Li_3PO_4 (space group Pnma).³² Lithium and phosphorus occupy the tetrahedral interstices, leaving all the octahedral sites empty, within the distorted hexagonal oxygen sublattice. All the LiO_4 tetrahedra point in the same direction in β - Li_3PO_4 while LiO_4 tetrahedra of γ - Li_3PO_4 point in the two opposite directions. c) Compositions in the solid solutions among Li_3PO_4 , Li_3VO_4 , and Li_4GeO_4 investigated in this study, $\text{Li}_3\text{P}_{1-x}\text{V}_x\text{O}_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1), $\text{Li}_{3+x}\text{V}_{1-x}\text{Ge}_x\text{O}_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1) and $\text{Li}_{3+x}\text{P}_{1-x}\text{Ge}_x\text{O}_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1).

Lithium-ion conducting solid-state electrolytes are a promising candidate to replace aprotic electrolytes currently used in lithium-ion batteries,¹⁻⁴ which can potentially allow the use of metallic lithium instead of graphite to markedly boost energy density stored and eliminate the electrolyte flammability to increase battery safety. Isovalent and/or aliovalent substitution of cations and anions in a number of structural families such as LISICON-like (lithium superionic conductor),⁵ garnets,⁶ NASICON-like (sodium superionic conductor),⁷ perovskites^{8,9} and lithium halides,¹⁰ can lead to significantly increase lithium ion conductivity,³ with the highest approaching that of liquids. Current major challenges in developing solid-state lithium-ion batteries rest on the instability^{2,3,5-17} between fast solid lithium-ion conductors and lithium-ion battery electrodes, highlighting critical needs to search for new solid-state lithium conductors and interfaces with high lithium-ion conductivity and stability. The vast majority of current research to increase lithium-ion conductivity, $\sigma T = \sigma_0 \exp(-E_a/k_B T)$, is directed towards lowering energetic barrier (E_a) for lithium ion migration in the structure. This can be done through creating tetrahedral pathways in the body centered cubic structure¹⁸ and/or employing larger and more polarizable anions (e.g. replacing oxygen with sulfur).^{3,19-21} However, lowering activation energy is shown to reduce the pre-exponential factor, $\sigma_0 = \sigma_{00} \exp(S_m/k_B)$, by several orders of magnitude,²¹⁻²⁵ where σ_{00} depends on jump frequency and mobile lithium-ion concentration and S_m is the entropy of migration. Generally speaking, large changes in σ_0 over many orders of magnitude most likely results from $\exp(S_m/k_B)$ as the variations in other parameters such as jumping frequency and mobile charge carrier concentration²⁶ are expected to fall within one order of magnitude^{21,23,25} for ion conductors with partial occupancy. For example, West et al.²² have shown that lowering the activation energy from 1.0 to 0.6 eV is accompanied with a decreased

pre-exponential factor, σ_0 , of LISICON $\text{Li}_{2+2x}\text{Zn}_{1-x}\text{GeO}_4$ solid solutions by ~ 6 orders of magnitude. This correlation is known as the Meyer-Neldel rule observed widely for a number of processes including bulk and surface diffusion and reaction kinetics.^{21-23,26-28} The correlation between σ_0 and E_a can be rationalized by invoking a linear relation $S_m = E_m/T_0$ where T_0 is related to the temperature of lithium sublattice disordering^{29,22} or $S_m/k_B = E_m/\Delta_0$ to yield Δ_0 , which is the energy scale of the excitation in the system according to the multi-excitation entropy theory,^{23,28} and represents the inverse of the slope in a Meyer-Neldel plot. As the entropy contribution to ion conductivity can overcome that from activation energy with increasing temperature, optimizing and rationally designing both migration barrier and the pre-exponential factor is needed to maximize ion conductivity. Here we take advantage of the structural flexibility of LISICONs and vast tunability of the ionic conductivity (~ 15 orders of magnitude^{3,5,20,30,31}) and activation energy (1.3³² to 0.21 eV²⁰) via cation and anion substitution in Li_3PO_4 . In the crystal structure of LISICON, lithium and phosphorus occupy the tetrahedral interstices, leaving all the octahedral sites empty within the distorted hexagonal oxygen sublattice, where all the LiO_4 tetrahedra can point in the same direction (space group $\text{Pmn}2_1$)³², as shown in Figure 1a, or LiO_4 tetrahedra can point in the two opposite directions (space group Pnma)³¹, as shown in Figure 1b. In this work, the lithium-ion conductivity has been investigated by a systematic substitution of lithium phosphate (Li_3PO_4 , with ionic radius of 0.17 Å for P^{5+}) with isovalent vanadium (ionic radius of 0.36 Å for V^{5+}) and with aliovalent germanium (ionic radius of 0.39 Å for Ge^{4+})³³, as shown in Figure 1c. We report that a decreasing activation energy is shown to correlate with reducing pre-exponential factor of lithium-ion conductivity of isovalent- and aliovalent-substituted Li_3PO_4 . The physical origin of this correlation is discussed in the context of

multi-phonon excitation entropy²⁸ and extended to other families of Li-ion conductors with or without partial occupancy.

2. METHODS

2.1 Synthesis and Experimental Characterizations.

LISICONs were synthesized through solid-state synthesis methods. Appropriate mixtures (5% excess lithium for Li_3PO_4 and $\text{Li}_{3.2}\text{Ge}_{0.2}\text{P}_{0.8}\text{O}_4$, and stoichiometric mixtures in all other cases) of Li_2CO_3 (99.998% Alfa Aesar), $(\text{NH}_4)_2\text{HPO}_4$ (98% Strem Chemicals), V_2O_5 (>99.6% trace metal basis, Aldrich) and GeO_2 (>99.99% trace metal basis, Aldrich) were ground with a mortar and pestle. The resulting powder mixtures were calcined in dry air at 800 °C for 10 hrs using a ramp rate of 5 °C/min for cooling and heating (for all synthesis) to produce the desired material. These powders were pressed at 1 GPa

into pellets with a 6 mm diameter and were 1-1.5 mm in thickness, which were then sintered at 800 °C for 10 hrs using a ramp rate of 5 °C/min. $\text{Li}_3\text{V}_{0.4}\text{P}_{0.6}\text{O}_4$ required an additional calcination at 900 °C to achieve a phase-pure solid solution, before the powders were sintered at the previously described sintering conditions. Sintered pellets were found to have porosities in the range of 7 to 33% (Table S1). Scanning electron microscope images of fractured pellets of Li_3PO_4 , Li_3VO_4 , Li_4GeO_4 , $\text{Li}_{3.4}\text{Ge}_{0.4}\text{P}_{0.6}\text{O}_4$, $\text{Li}_{3.4}\text{Ge}_{0.4}\text{V}_{0.6}\text{O}_4$, and $\text{Li}_3\text{V}_{0.4}\text{P}_{0.6}\text{O}_4$ were taken. Li_3PO_4 and $\text{Li}_{3.4}\text{Ge}_{0.4}\text{P}_{0.6}\text{O}_4$ showed little porosity with grain sizes of 10-20 μm , which are in agreement with the porosity measurements of 7% and 9%, respectively. Li_3VO_4 , Li_4GeO_4 , $\text{Li}_{3.4}\text{Ge}_{0.4}\text{V}_{0.6}\text{O}_4$, and $\text{Li}_3\text{V}_{0.4}\text{P}_{0.6}\text{O}_4$ images showed some secondary morphology with nano-porosity, which agrees with the higher porosities of 13%, 24%, 17%, and 13% found, respectively.

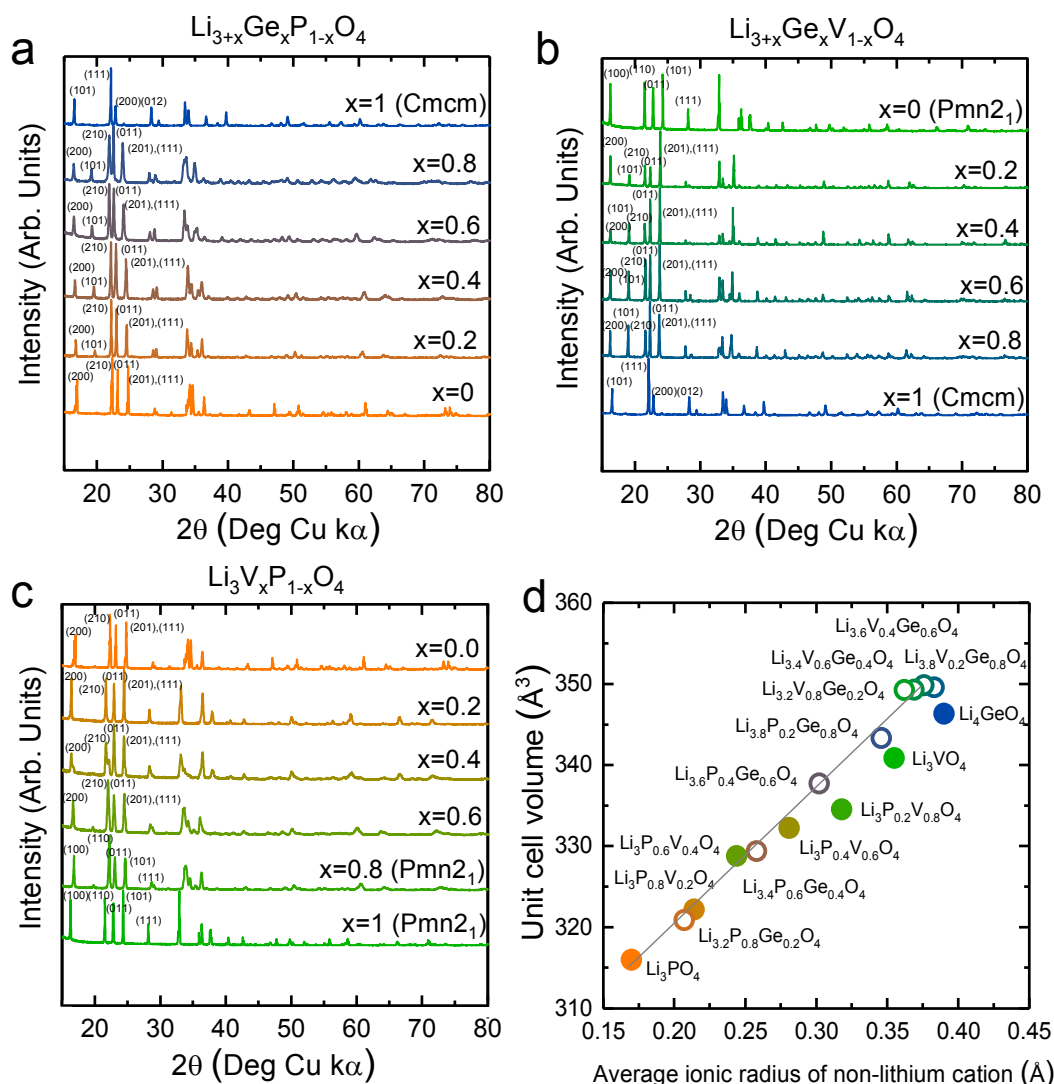


Figure 2. X-ray diffraction patterns of LISICON solid solutions of a) $\text{Li}_{3+x}\text{P}_{1-x}\text{Ge}_x\text{O}_4$, b) $\text{Li}_{3+x}\text{V}_{1-x}\text{Ge}_x\text{O}_4$ and c) $\text{Li}_3\text{P}_{1-x}\text{V}_x\text{O}_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1). All the compounds crystallize into Pnma space group with the exception of Li_3VO_4 and $\text{Li}_3\text{V}_{0.8}\text{P}_{0.2}\text{O}_4$ which has $\text{Pmn}2_1$ symmetry and Li_4GeO_4 which has Cmcm symmetry. d) Fitted unit cell volume of compounds in a, b, c as a function of average non-lithium cation sizes (P^{5+} , V^{5+} and Ge^{4+}) in the crystal structure. The lattice parameters and unit cell volumes were refined with Pnma symmetry, unless otherwise noted in parentheses, and are listed in Table S2 and are in good agreement with previously published results.⁵ The unit cell volume for compounds with $\text{Pmn}2_1$ symmetry were multiplied by two as the number of formula units is two in $\text{Pmn}2_1$ and four in Pnma and Cmcm . The average radii were computed by adding the Shannon radius³³ of each non-lithium cation weighted by their corresponding fraction in the each sample.

The phase purity of LISICON sintered pellet samples were confirmed using X-ray diffraction on a Panalytical X'Pert Pro diffractometer with Cu $\text{K}\alpha$ radiation with one-hour continuous scans from 15–80 degrees 2θ . Lattice parameters of all samples were refined using profile matching of Fullprof based on space group Pnma except for Li_3VO_4 and $\text{Li}_3\text{V}_{0.8}\text{P}_{0.2}\text{O}_4$ having $\text{Pmn}2_1$ symmetry, and Cmcm symmetry for Li_4GeO_4 (Table S2). Pseudo-Voigt function was used for peak shape to refine the peak positions in order to determine the lattice parameters. The atomic positions and occupancy were not refined. An example of profile-matching-fitted and experimental patterns is shown in Figure S2. Elemental analyses of cations were determined using inductively coupled plasma-optical emission spectroscopy (ICP-AES) with an Agilent 5100 ICP-AES, which showed a good agreement with targeted stoichiometries

(Table S3). Samples and references were dissolved in nitric acid to be measured. The porosity of sintered pellets was determined by measurement with a micrometer, which was consistent with scanning electron microscope (SEM) secondary electron images of fractured cross-sections of pellets on a Zeiss Merlin SEM.

Pellets were sputtered with 100 nm of gold on each face and were generally stored within an argon-filled glove box to avoid reactions with water and carbon dioxide. Electrochemical impedance spectroscopy (EIS) data were collected from gold sputtered pellets assembled between two stainless steel current collectors with Biologic potentiostats with VSP300 boards. Data were collected between 7 MHz and 0.1 Hz using an AC voltage amplitude 1 V to increase the signal to noise

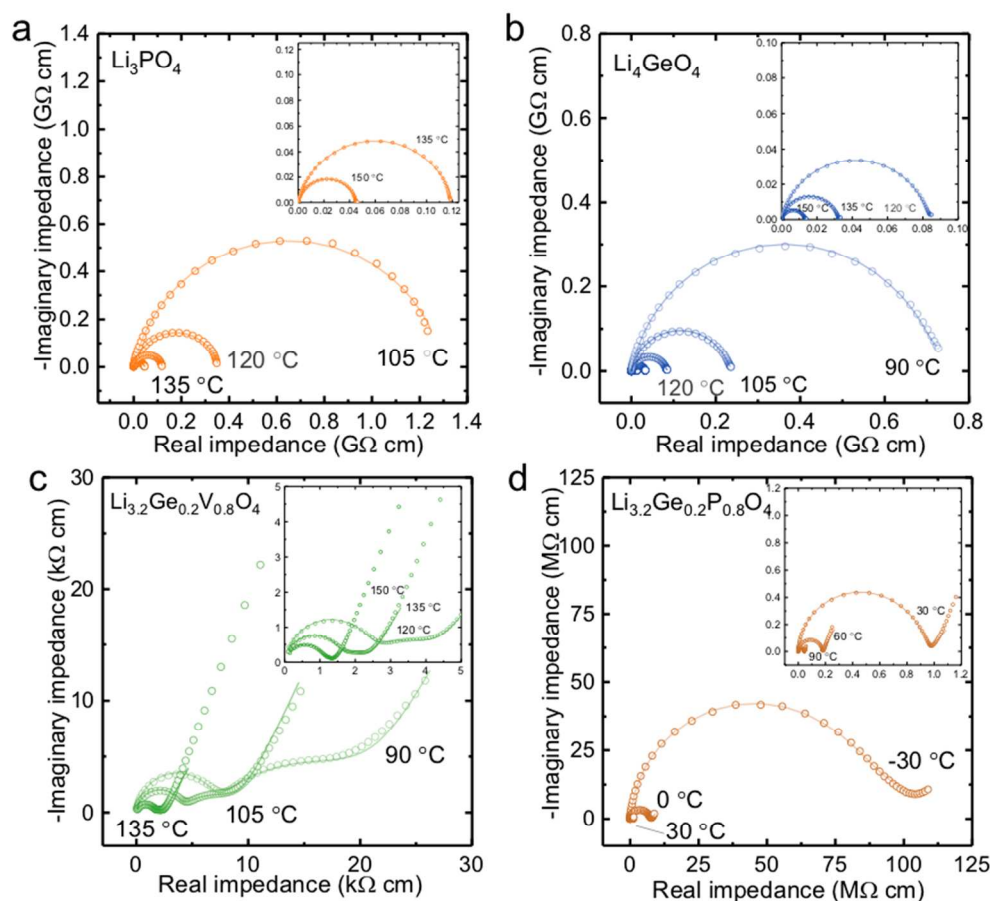


Figure 3. Nyquist impedance plots of a) Li_3PO_4 , b) Li_4GeO_4 , c) $\text{Li}_{3.2}\text{Ge}_{0.2}\text{V}_{0.8}\text{O}_4$, and d) $\text{Li}_{3.2}\text{Ge}_{0.2}\text{P}_{0.8}\text{O}_4$ at select temperatures. The EIS spectra of compounds without partial lithium occupancy were fitted with one semi-circle using the equivalent circuit shown in Figure S3a while the EIS spectra of compounds with partial lithium occupancy were fitted with two semi-circle using the equivalent circuit shown in Figure S3b. Two semicircles of $\text{Li}_{3.2}\text{Ge}_{0.2}\text{P}_{0.8}\text{O}_4$ found at low temperatures merged into one semicircle at temperatures greater than 30°C in Figure 3d. The open circles are the collected data points, while the lines joining the open circles correspond to the fits with the employed equivalent circuits.

ratio for measurements of highly resistive samples. All reported values are from experiments with an AC voltage amplitude of 1 V, however results were compared to measurement with 10 mV AC amplitude on low resistivity and samples and were found to be in good agreement. The temperature of the samples during EIS measurements was controlled with an Espec SJ-241 environmental chamber between -30°C and 150°C . Impedance spectra were fit to the equivalent circuit using Z-fit within EC-Lab from Bio-Logic, from which the extracted high-frequency semi-circle resistance was used to compute lithium-ion resistivity as a function of temperature. Impedance results were fitted with resistance and constant-phase elements (Q) as $R_{\text{series}} + R_{\text{bulk}}/Q_{\text{bulk}} + Q_{\text{elec}}$ for results with one distinct semi-circle using the equivalent circuit (Figure S3a) and $R_{\text{series}} + R_{\text{bulk}}/Q_{\text{bulk}} + R_{\text{gb}}/Q_{\text{gb}} + Q_{\text{elec}}$ for two distinct semicircles using the equivalent circuit (Figure S3b). R_{series} represents the resistance of the leads and cell, Q_{elec} represents the imperfect capacitance of the planar electrode surface at low frequencies, $R_{\text{bulk}}/Q_{\text{bulk}}$ represent the parallel resistance and capacitance of the bulk material, and $R_{\text{gb}}/Q_{\text{gb}}$ represents the parallel resistance and capacitance of the grain boundaries.⁵

2.2 Classical Molecular Dynamics (MD) simulations

Classical molecular dynamics (MD) simulations were performed using LAMMPS package.³⁴ The interatomic potential used in this work was fitted to reproduce the structural and mechanical properties of phospho-silicate materials^{35,36} and was shown to give reasonably good accuracy in term of ionic conductivity on chemically similar systems ($\text{Li}_{4+x}\text{Si}_{1-x}\text{M}_x\text{O}_4$ ($\text{M} = \text{P}, \text{Al}$ or Ge)³⁰ and Li_4SiO_4 - Li_3PO_4 ³⁷) as well as in our Li_3PO_4 - Li_3VO_4 - Li_4GeO_4 system as can be seen from the good agreement between computed migration barriers and measured activation energies that we will show in later section. The potential contains three terms: a Coulomb term to account for the long-range interaction between ions, a Morse term for the short-range repulsive interaction and a r^{-6} term which was found to be important to accurately model the system especially in the high-temperature regime by making the repulsive interaction 'harder'. We used a $4 \times 4 \times 2$ supercell of Li_3VO_4 and Li_3PO_4 (Pnma) and a $4 \times 4 \times 4$ supercell of Li_4GeO_4 containing respectively 1024 and 1152 atoms. We did not perform a convergence study with respect to the size of the supercell because previous computational study of the ionic conductivity of $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ system¹⁹, the diffusivity already converges when only one unit cell was used. Therefore, we expect that our results are well converged with respect to the size of the

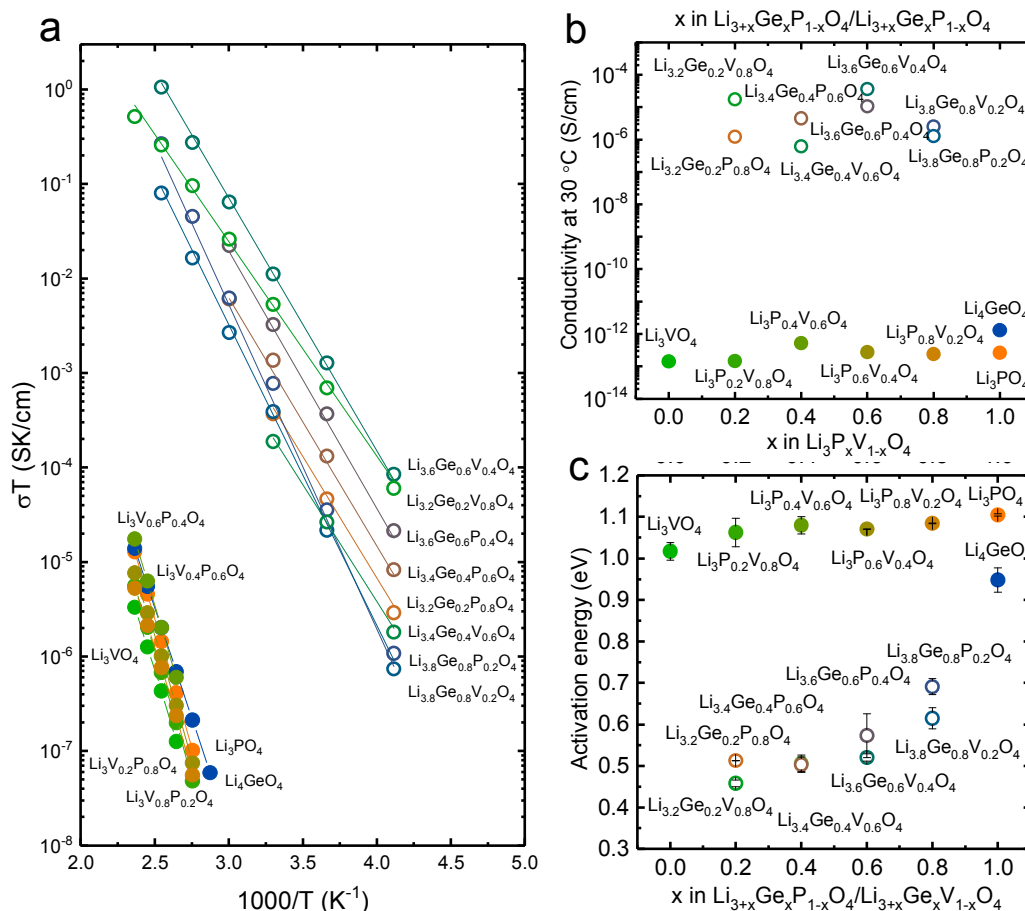


Figure 4. a). Arrhenius plot of $Li_3P_{1-x}V_xO_4$, $Li_{3+x}V_{1-x}Ge_xO_4$ and $Li_{3+x}P_{1-x}Ge_xO_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1). For compounds with partial Li occupancy, the conductivity reported here are bulk conductivity. b) Conductivity at 30 °C and c) Activation energy of $Li_{3+x}P_{1-x}Ge_xO_4$, $Li_{3+x}V_{1-x}Ge_xO_4$, and $Li_3P_{1-x}V_xO_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1). The conductivity of all compound without partial Li occupancy were extrapolated from high-temperature measurements.

supercell used in this study. We also simulated non-stoichiometric compositions $Li_{3+x}V_{1-x}Ge_xO_4$ ($x = 0.2, 0.4, 0.6$ and 0.8) by randomly replacing V by Ge in the supercell and introducing interstitial lithium to make the whole supercell electrically neutral. The supercells were relaxed until the forces on the ions fall below 10^{-6} eV/Å. The MD simulations were performed in NVT ensemble using Nose-Hoover thermostat to control the temperature while the volumes of the unit cells were fixed at their experimental values. A step of 2 fs was used for all simulations. For compounds $Li_{3+x}Ge_xV_{1-x}O_4$ ($x = 0.2, 0.4, 0.6$ and 0.8), the simulations were done at 4 different temperatures: 770K for 150ps, 870K for 125ps, 1000K for 100ps and 1250K for 100ps. For Li_4GeO_4 , the simulation was performed at 870 K for 1000 ps, 1000 K for 750 ps, 1250 K for 500 ps and 1500 K for 500 ps. We found that the diffusivities are well-converged with these simulation times as can be seen from the linear increase of mean-square displacements (MSD) as function of time as shown in Figure S4. The (self) diffusion coefficient was extracted from the slope of the MSD versus. The lithium-ion density and Van-Hove correlation function were computed from the MD trajectories using the pymatgen_diffusion module in the Pymatgen package.^{38,39} The computed lithium-ion density and Van-Hove correlation functions are shown in Figure S5 and S6 and discussed in detail in the SI.

3. RESULTS

3.1 Structural analyses of $Li_3P_{1-x}V_xO_4$, $Li_{3+x}V_{1-x}Ge_xO_4$ and $Li_{3+x}P_{1-x}Ge_xO_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1)

X-ray powder diffraction analysis of $Li_{3+x}P_{1-x}Ge_xO_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1) in Figure 2a, $Li_{3+x}V_{1-x}Ge_xO_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1) in Figure 2b and $Li_3P_{1-x}V_xO_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1) in Figure 2c reveal that these samples were phase pure, where major diffraction reflections were indexed to the previously reported crystal structures and space groups.³² The lattice parameters of $Li_3P_{1-x}V_xO_4$, $Li_{3+x}V_{1-x}Ge_xO_4$ and $Li_{3+x}P_{1-x}Ge_xO_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1) were obtained from profile matching using Fullprof (Table S2) and are found consistent with previous work.^{5,32} Of significance to note is that the unit cell volumes of the LISICON structures $Li_3P_{1-x}V_xO_4$ and $Li_{3+x}P_{1-x}Ge_xO_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1) were found to increase with Ge^{4+} and V^{5+} content (Figure 2d) reflecting the larger ionic size³³ of Ge^{4+} and V^{5+} (0.39 Å and 0.35 Å) compared to P^{5+} (0.17 Å). In contrast, the unit cell volumes of $Li_{3+x}V_{1-x}Ge_xO_4$ were found to vary little with Ge^{4+} or V^{5+} content as expected from their similar ionic radii.

3.2 Lithium Ion conductivity of $Li_3P_{1-x}V_xO_4$, $Li_{3+x}V_{1-x}Ge_xO_4$ and $Li_{3+x}P_{1-x}Ge_xO_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1)

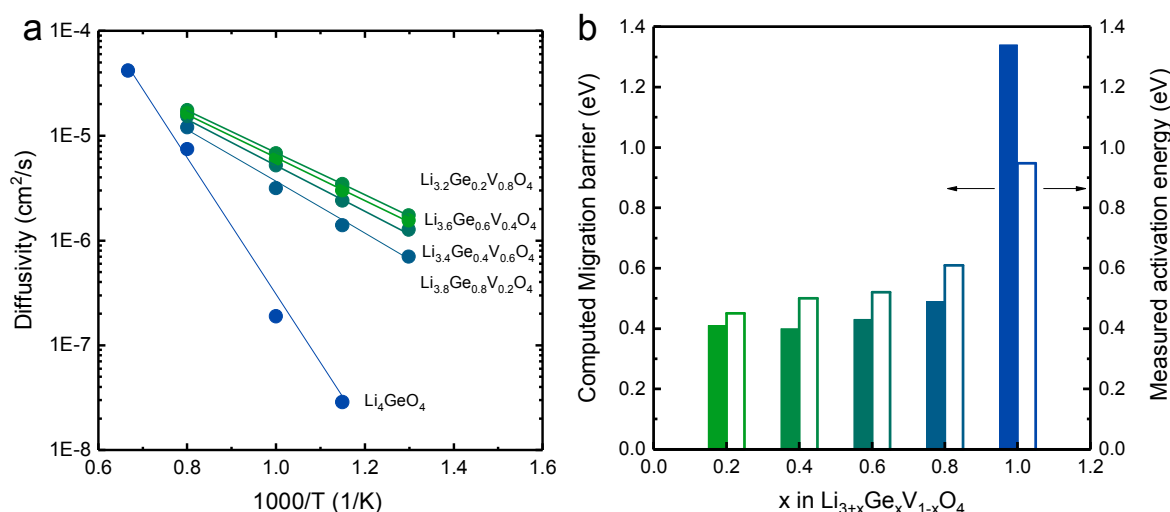


Figure 5. a) The computed (self) diffusion coefficient of $\text{Li}_{3+x}\text{Ge}_x\text{V}_{1-x}\text{O}_4$ ($x = 0.2, 0.4, 0.6, 0.8$ and 1). b) Comparison between the computed migration barrier and measured activation energy. While the computed migration barrier was slightly lower than the experimental values, the migration barrier for lithium-ion diffusion was found to become greater with increasing Ge content which resembles the experimental trend shown in Figure 4c. The computed diffusivities are well converged as can be seen from the linear increase of MSD versus time shown in Figure S4.

For LISICONs without partial lithium occupancy ($\text{Li}_3\text{P}_{1-x}\text{V}_x\text{O}_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1) and Li_4GeO_4), the impedances were measured over the temperature range 90°C to 150°C (below 90°C the resistances were too high). For compounds with partial lithium occupancy ($\text{Li}_{3+x}\text{V}_{1-x}\text{Ge}_x\text{O}_4$ and $\text{Li}_{3+x}\text{P}_{1-x}\text{Ge}_x\text{O}_4$ ($x = 0.2, 0.4, 0.6$ and 0.8)), the impedances were measured over the temperature range -30°C to 150°C . LISICONs without partial lithium occupancy exhibited only one semicircle while compounds with partial lithium occupancy exhibited two distinct semi-circles below $\sim 30^\circ\text{C}$ - 60°C which can be attributed to the bulk and grain boundary conductivity. Nyquist plots of Li_3PO_4 and Li_4GeO_4 are shown in Figure 3a and 3b, respectively. Lithium-ion conductivity of these conductors estimated from the semicircle Li_4GeO_4 and $\text{Li}_3\text{P}_{1-x}\text{V}_x\text{O}_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1) using the equivalent circuit shown in Figure S3a, yield values of $\sim 2 \times 10^{-10}$ S/cm for Li_3PO_4 and $\sim 2 \times 10^{-9}$ S/cm for Li_4GeO_4 at 100°C , which are consistent with previous reports.^{40,41} In addition, these conductors were found to have high activation energy of ~ 1.1 eV (Figure 4), in agreement with previous studies,^{40,41} which can be attributed to both enthalpy of migration and defect formation energy. Isovalent substitution of P^{5+} for V^{5+} in the crystal structure (space group $Pnma$) did not lead to any noticeable changes in the ion conductivity nor activation energy as expected. In contrast, aliovalent substitution of P^{5+} or V^{5+} for Ge^{4+} in Li_3PO_4 and Li_3VO_4 creates partial lithium occupancy and results in much higher conductivities and lower activation energies, as shown in Figure 4. EIS data were fitted with two semicircles in Nyquist plots using the equivalent circuit (Figure S3b). Examples of $\text{Li}_{3.2}\text{Ge}_{0.2}\text{P}_{0.8}\text{O}_4$ and $\text{Li}_{3.2}\text{Ge}_{0.2}\text{V}_{0.8}\text{O}_4$ are shown in Figure 3c and 3d, respectively. Two semicircles of $\text{Li}_{3.2}\text{Ge}_{0.2}\text{P}_{0.8}\text{O}_4$ found at low temperatures merged into one semicircle at temperatures greater than 30°C in Figure 3d. The high-frequency semicircle was attributed to bulk conduction, and the low-frequency semicircle was attributed to grain boundary conduction as reported previously.⁵ The ~ 45 -degree tail at low frequencies was assigned to effects of the planar

gold electrode surface.⁵ The Bode plots of all compositions at various temperatures are shown in Figure S7. The capacitance as well as the ideality factor of the constant phase elements are shown in Table S4. For compounds with partial lithium occupancy the capacitance associated with the grain boundary is larger than that of the bulk, as expected.⁴² For compounds without partial lithium occupancy, only one semi-circle can be resolved, but based on their capacitances, we can conclude that the main contribution to this total capacitance come from the diffusion process that occurs in the bulk instead of the grain boundary.

Below we discuss the trends in the lithium-ion conductivity and activation energy found in the $\text{Li}_{3+x}\text{P}_{1-x}\text{Ge}_x\text{O}_4$ and $\text{Li}_{3+x}\text{V}_{1-x}\text{Ge}_x\text{O}_4$ with partial lithium occupancy. Aliovalent substitution of P^{5+} or V^{5+} by Ge^{4+} by 20% led to marked increase in lithium-ion conductivity by 4-5 orders of magnitude relative to Li_3PO_4 and Li_3VO_4 in Figure 4b. This increase is accompanied with a large reduction in the activation energy of ~ 0.5 eV in Figure 4c, which can be explained by the reduction of formation energy of the mobile charge carrier in the activation energy for compounds with partial lithium occupancy. Further increasing the amount of Ge^{4+} substitution in Li_3PO_4 and Li_3VO_4 did not result in any significant changes in the lithium-ion conductivity (Figure 4b) while the activation energy was found to increase gradually with greater Ge substitution in Figure 4c, which can be attributed to increasing migration barrier with increasing Ge^{4+} in the crystal structure in agreement with MD results (Figure 5a and 5b) which also shows that the computed migration barrier for lithium diffusion increases with Ge substitution in $\text{Li}_{3+x}\text{Ge}_x\text{V}_{1-x}\text{O}_4$. This trend cannot be explained by the variation in the volume of the unit cell as can be seen from Figure S8b, the unit cell volume of the $\text{Li}_{3+x}\text{V}_{1-x}\text{Ge}_x\text{O}_4$ series barely changes while that of the $\text{Li}_{3+x}\text{P}_{1-x}\text{Ge}_x\text{O}_4$ series increases with Ge content. We hypothesize that this trend is due to the inductive effect by which Ge^{4+} , which is less positively charged than P^{5+} or V^{5+} , pulls less electron den-

sity from oxygen resulting in higher effective charge on oxygen which in turns leads to stronger electrostatic interaction with the lithium ions and higher activation energies. Because the empirical potential we used for the classical MD simulations cannot capture the inductive effect, more advanced computational techniques such as *ab-initio* MD are needed to support this hypothesis. This trend is also consistent with variation in activation energy and pre-factor found in the lithium superionic conductor $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$ series.⁴³ It should be noted that the computed migration barrier for lithium diffusion was slightly smaller than the measured activation energy for compositions with partial occupancy while the computed migration barrier of Li_4GeO_4 was higher than the experimental value. Many factors including differences in the stoichiometry, unit cell sizes, the use of empirical potential and the absence of grain boundary and porosity in the MD simulation can contribute to this discrepancy. While the variation of the unit cell volume can be excluded as one possible explanation for the observed trend of activation energy as a function of Ge content, we cannot entirely exclude subtler structural changes such as polyhedral volumes or the size of the ‘diffusion pathway’ which require a more detailed analysis of atomic positions in the unit cell.

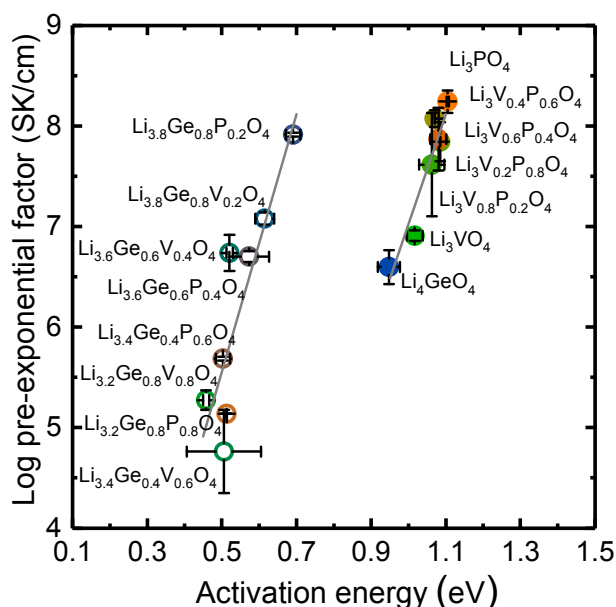


Figure 6. Log of pre-exponential factor σ_0 from $\sigma T = \sigma_0 \exp(-E_a/kT)$ versus activation energy, E_a . Error bars show standard deviation of 2-3 samples for each composition. The pre-exponential factor and activation energy for all samples studied are shown in Table S4.

The pre-exponential factor (σ_0) of lithium ion conductivity for all the LISICON conductors with full and partial occupancy was found to scale with the activation energy (E_a) as reported previously,^{21–27} as shown in Figure 6 and Table S5. We found this correlation also applies to the pre-exponential factor of diffusivity D_0 and the migration barrier E_m extracted from MD simulation as can be seen in Figure S9 in the SI. This correlation, also known as Meyer-Neldel rule can be explained within the framework of multi-excitation entropy (MEE), proposed by Yelon et al.,²⁸ based on the idea that the activation entropy (in the diffusion process, this is the entropy of migration S_m)

arises from the multiplicity of microstates (in a statistical sense) associated with ways in which excitations (phonons) are to be annihilated to provide enough energy to the system to overcome the (migration) barrier E_m . Within this theory, the slope of $\log(\sigma_0)$ versus E_a is inversely proportional to the energy scale of the excitations (phonons) Δ_0 which is on the order of ~ 10 -100 meV. While MEE can explain the correlation between σ_0 and E_a , it does not provide an explanation for the difference of the intercepts between stoichiometric and non-stoichiometric compounds in Figure 6. In this paper, we show that this can be explained in terms of the Gibbs free energy of formation of mobile charge carrier G_f . The derivation is based on the microscopic model of diffusion⁴⁴ and detailed in the SI. The main result from this derivation is the equation below:

$$\ln(\sigma_0) = \ln(\sigma_{00}) - \frac{G_f(T_0)}{2\Delta_0} + \frac{E_a}{\Delta_0} \quad (1)$$

Where $T_0 \equiv \Delta_0/k_B$. The first term on the right-hand side contains various microscopic parameter such as attempt frequency, jump distance, number of nearest neighbors, Haven ratio, correlation factor, maximum concentration of mobile species (please refer to the SI for more detail) which can be estimated and is not expected to vary by several orders of magnitude for different materials while G_f is known to have much larger variations for stoichiometric compounds compared non-stoichiometric compounds and accounts for the difference in the intercepts in Figure 6. The values of Δ_0 for the series of compounds with and without partial lithium occupancy are 33.5 meV and 39.5 meV, respectively. These values are very similar, reflecting the closeness in ‘phonon energy’ in these two chemically similar series of compounds.⁴⁵ Meyer-Neldel’s rule can also be found in other lithium-ion conductors such as LISICON $\text{Li}_{4-2x}\text{Zn}_x\text{GeO}_4$ (Figure S10a), thio-Lisicon $\text{Li}_{3+x}\text{Ge}_x\text{P}_{1-x}\text{S}_4$ (Figure S10b), $\text{Li}_{3+x}\text{Ti}_x\text{As}_{1-x}\text{O}_4$ (Figure S10c) and garnet $\text{Li}_5\text{La}_3\text{Nb}_{2-x}\text{Y}_x\text{O}_{12}$ (Figure S10d). Δ_0 in these materials were found to be ~ 15 - 60 meV which correspond well to the energy scale of phonon in solids. To further quantify this phonon energy scale, we use the (computed) Debye temperature⁴⁶ as well as phonon band centers which were recently proposed as descriptor for enthalpy of migration in LISICON and Olivine compounds.⁴⁵ As seen from Figure S11, Δ_0 seems to decrease with increasing phonon band center and Debye temperature. Obviously, more data are needed to confirm this trend whose physical origin is still an open question.

We have also estimated the Gibbs free energy of mobile carrier formation in some lithium-ion conductors from the intercept with the y-axis using equation (1) and obtained reasonable values ~ 0.5 - 2 eV (Figure S12a), which is in agreement with previous work. For example, the formation energy of Frenkel defects in $\gamma\text{-Li}_3\text{PO}_4$ was found to be ~ 1.7 eV from previous computational study.⁴⁷ Moreover, as expected, the Gibbs free energy of defect formation decreases upon aliovalent substitution, as shown in Figure S12b for LISICONs and garnets. Finally, we have also compiled the values of pre-exponential factors and activation energies of several compounds from the literature and plotted them in Figure 7a (the actual data and references can be found in Table S6). Although within each family of compounds, the data follow Meyer-Nedel’s rule, their slopes and intercepts are different, reflecting the difference in the phonon energy scale and the Gibbs free energy of defect formation in these materials. However, based on equa-

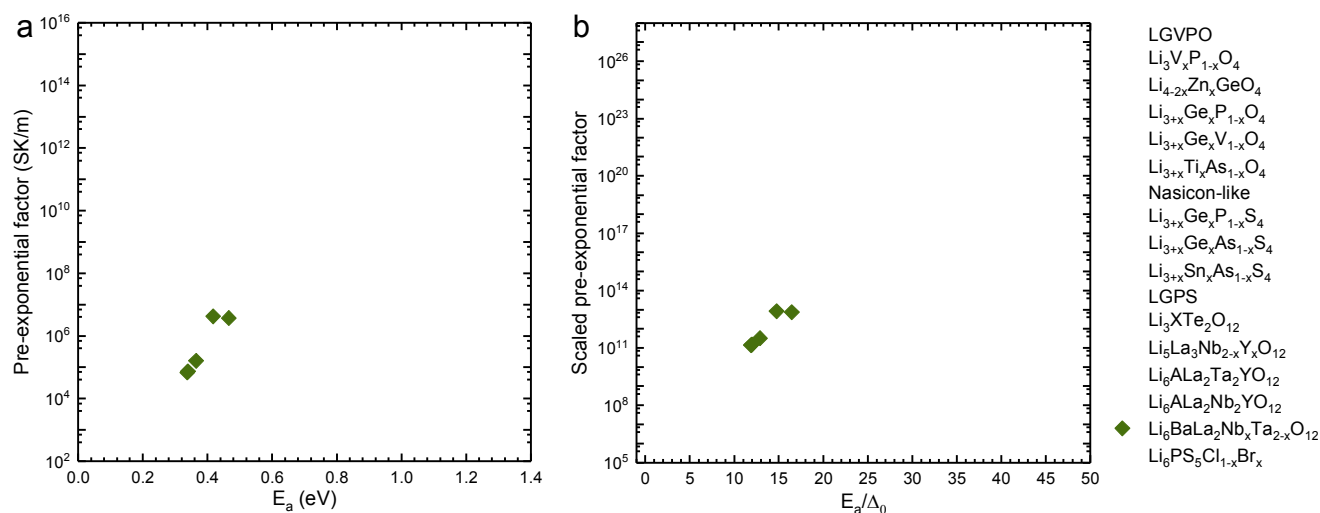


Figure 7 a) Plot of pre-exponential factor versus activation energy for several families of lithium-ion conductors. Different families of lithium-ion conductors exhibit different slopes and intercepts reflecting the difference in their phonon energy and Gibbs free energy of defect formation. b) After rescaling the pre-exponential factor and activation energy using equation (1), all the points collapse on a single straight line with the slope of 1 as expected. The reference for the data of each series compound included here are $\text{Li}_{4-2x}\text{Zn}_x\text{GeO}_4$,⁴⁸ $\text{Li}_{3+x}\text{V}_{1-x}\text{Ge}_x\text{O}_4$,⁵ $\text{Li}_{3+x}\text{As}_{1-x}\text{Ti}_x\text{O}_4$,⁵ Nasicon-like $\text{LiX}(\text{PO}_4)_3$,^{7,49} $\text{Li}_{4-x}\text{Ge}_{1-x}\text{P}_x\text{S}_4$,²⁰ $\text{Li}_{3+x}\text{Ge}_x\text{As}_{1-x}\text{S}_4$,⁵⁰ $\text{Li}_{4-x}\text{Sn}_{1-x}\text{As}_x\text{S}_4$,⁵¹ $\text{Li}_{10}\text{Ge}_{1-x}\text{Sn}_x\text{P}_2\text{S}_{12}$,⁵² $\text{Li}_{10}\text{Ge}_{1-x}\text{Si}_x\text{P}_2\text{S}_{12}$,⁵² $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$,⁵³ $\text{Li}_3\text{X}_3\text{Te}_2\text{O}_{12}$,⁵⁴ $\text{Li}_5\text{La}_3\text{Nb}_{2-x}\text{Y}_x\text{O}_{12}$,⁵⁵ $\text{Li}_6\text{Ala}_2\text{Ta}_2\text{O}_{12}$,⁵⁶ $\text{Li}_6\text{Ala}_2\text{Nb}_2\text{O}_{12}$,⁵⁷ $\text{Li}_6\text{BaLa}_2\text{Nb}_x\text{Ta}_{2-x}\text{O}_{12}$,⁵⁸ and $\text{Li}_6\text{PS}_5\text{Cl}_{1-x}\text{Br}_x$.²¹ The actual data can be found in Table S5.

tion (1), it is possible to rescale this data so that they collapse on the same straight line by normalizing E_a by Δ_0 and switching the second term, containing G_f , to the left-hand side. The rescaled relationship is shown in Figure 7b, and all compounds indeed fall on the same straight line with the expected slope of 1, confirming the validity of Meyer-Neldel's rule and the consistency of our analysis. Notice that in this analysis, G_f and Δ_0 were both extracted from Meyer-Neldel plot making the scaling in Figure 7b rather trivial. However, if one can get the values of G_f or Δ_0 from separate measurements or computations, this scaling analysis would provide much stronger evidence for the universality of Meyer-Neldel rule and its physical origin in terms of multi-excitation entropy.

CONCLUSIONS

In this study, the ionic conductivity and activation energy of lithium in the Li_3PO_4 - Li_3VO_4 - Li_4GeO_4 system was systematically investigated. It was found that doping Li_3PO_4 or Li_3VO_4 with Ge to create partial lithium occupancy in the system leads to an increase in room-temperature lithium conductivity by ~ 7 order of magnitude and a sharp decrease in the activation energy, which was attributed to the decrease in the defect formation energy. We explained the variation of activation energy as a function of Ge content by invoking the inductive effect which alters the electron density of the oxygen due to the difference in electronegativity of Ge^{4+} and P^{5+} or V^{5+} . It has also been found that the pre-exponential factor and activation energy are coupled in our samples in agreement with Meyer-Neldel rule and that the samples with partial occupancy lie on a separate line than those without partial occupancy, but they have a similar slope when plotting the pre-exponential Arrhenius factor versus the activation energy. We related these slopes to the energy scale of phonons in the system, according to the multi-excitation entropy theory.²⁸ The intercept of the correlation were related to the Gibbs free energy of defect for-

mation. A compilation of the pre-exponential factors and activation energy data for several commonly studied lithium-ion conductors showed this correlation is very general, therefore imposing an unfavorable trade-off between low activation energy and high pre-exponential factor required to reach high ionic conductivity. Nevertheless, it was found that in some systems such as β -alumina, the pre-exponential factors versus activation energies of different mobile species (Ag, Na, K and Tl) do not seem to follow the Meyer-Neldel rule.^{27,28} Understanding the physical origin behind Meyer-Neldel rule represents an exciting fundamental research opportunity and can also lead to new strategies to increase ionic conductivity by finding the conditions under which this correlation can experimentally be violated.

ASSOCIATED CONTENT

AUTHOR INFORMATION

Corresponding Author

*John.Bachman@calstatela.edu and shaohorn@mit.edu

Present Addresses

† Department of Mechanical Engineering, California State University Los Angeles, Los Angeles, CA, 90032.

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript. ‡These authors contributed equally

Notes

The authors declare no competing financial interests.

ACKNOWLEDGMENT

J.B. was supported by the National Science Foundation Graduate Research Fellowship Program under Grant No. 1122374. H.-H.C. was in part supported from the Ministry of Science and Technology of Taiwan (102-2917-I-564-006-A1). Research made use of facilities supported by the MRSEC Program of the National Science Foundation under award No. DMR – 0819762. Research funding and support was provided by BMW.

SUPPORTING INFORMATION

Figure S1: Scanning electron microscope images of fractured pellets of Li_3PO_4 , Li_3VO_4 , Li_4GeO_4 , $\text{Li}_{3.4}\text{Ge}_{0.4}\text{P}_{0.6}\text{O}_4$, $\text{Li}_{3.4}\text{Ge}_{0.4}\text{V}_{0.6}\text{O}_4$, and $\text{Li}_3\text{V}_{0.4}\text{P}_{0.6}\text{O}_4$.

Figure S2: Fitted lattice parameters of $\text{Li}_{3.2}\text{Ge}_{0.2}\text{V}_{0.8}\text{O}_4$ with goodness of fit

Figure S3: Equivalent circuit used to extract the value of resistance from EIS spectra

Figure S4: MSD displacement versus time obtained from MD simulations

Figure S5: Isosurface of Li^+ probability distribution function in $\text{Li}_{3.6}\text{Ge}_{0.6}\text{V}_{0.4}\text{O}_4$ at 1000 K calculated from the MD trajectories

Figure S6: Self and distinct parts of the Van Hove correlation function of $\text{Li}_{3.6}\text{Ge}_{0.6}\text{V}_{0.4}\text{O}_4$ and Li_4GeO_4 at 1000 K

Figure S7: Bode plots of $\text{Li}_{3-x}\text{V}_x\text{O}_4$, $\text{Li}_{3+x}\text{V}_{1-x}\text{Ge}_x\text{O}_4$ and $\text{Li}_{3+x}\text{P}_{1-x}\text{Ge}_x\text{O}_4$ ($x = 0, 0.2, 0.4, 0.6, 0.8$ and 1)

Figure S8: Average conductivity at 105 °C, average activation energy as a function of lattice volume and average activation energy as a function of porosity

Figure S9: Pre-exponential factor of the diffusivity D_0 as a function of the migration barrier E_m extracted from MD simulations.

Figure S10: Plot of pre-exponential factor versus activation

Figure S11: Correlation of the phonon energy scale Δ_0 as estimated from the inverse of the slope of the Meyer-Neldel plot as functions of phonon band and computed Debye temperature

Figure S12: Gibbs free energy of formation of mobile charge carrier estimated from the intercept of Meyer-Neldel plots

Table S1: Average porosity of two or three samples measured by micrometer and compared to unit cell size refined from XRD measurements.

Table S2: Lattice parameters obtained from profile matching with X-ray diffraction

Table S3: Atomic ratios between cations measured from inductively couple plasma-atomic emission spectroscopy

Table S4: Capacitance and the ideality factor α of the constant phase element

Table S5: Log (base ten) of pre-exponential Arrhenius factor and activation energy for all samples, and average and standard deviation for those samples

Table S6. Compiled data of pre-exponential factors and activation energies

Derivation of equation (1) relating σ_0 to E_a

REFERENCES

- (1) Goodenough, J. B.; Kim, Y. Challenges for Rechargeable Li Batteries. *Chem. Mater.* **2010**, *22* (3), 587–603.
- (2) Janek, J.; Zeier, W. G. A Solid Future for Battery Development. *Nat. Energy* **2016**, *1* (9), 16141.
- (3) Bachman, J. C.; Muy, S.; Grimaud, A.; Chang, H.-H.; Pour, N.; Lux, S. F.; Paschos, O.; Maglia, F.; Lupart, S.; Lamp, P.; et al. Inorganic Solid-State Electrolytes for Lithium Batteries: Mechanisms and Properties Governing Ion Conduction. *Chem. Rev.* **2016**, *116* (1), 140–162.
- (4) Lotsch, B. V.; Maier, J. Relevance of Solid Electrolytes for Lithium-Based Batteries: A Realistic View. *J. Electroceram* **2017**, *38* (2–4), 128–141.
- (5) Rodger, A.; Kuwano, J.; West, A. Li^+ Ion Conducting Gamma-Solid Solutions in the Systems $\text{Li}_4\text{XO}_4\text{-Li}_3\text{YO}_4$ - $\text{X}=\text{Si, Ge, Ti}$; $\text{Y}=\text{P, As, V}$; $\text{Li}_4\text{XO}_4\text{-Li}_2\text{ZO}_2\text{-Z}=\text{Al, Ga, Cr}$ and $\text{Li}_4\text{GeO}_4\text{-Li}_2\text{CaGeO}_4$. *Solid State Ionics* **1985**, *15* (3), 185–198.
- (6) Thangadurai, V.; Narayanan, S.; Pinzaru, D. Garnet-Type Solid-State Fast Li Ion Conductors for Li Batteries: Critical Review. *Chem. Soc. Rev.* **2014**, *43* (13), 4714–4727.
- (7) Martínez-Juárez, A.; Pecharrmán, C.; Iglesias, J. E.; Rojo, J. M. Relationship between Activation Energy and Bottleneck Size for Li^+ Ion Conduction in NASICON Materials of Composition $\text{LiMM}'(\text{PO}_4)_3$; $\text{M, M}' = \text{Ge, Ti, Sn, Hf}$. *J. Phys. Chem. B* **1998**, *102* (2), 372–375.
- (8) Bohnke, O.; Bohnke, C.; Fourquet, J. L. Mechanism of Ionic Conduction and Electrochemical Intercalation of Lithium into the Perovskite Lanthanum Lithium Titanate. *Solid State Ionics* **1996**, *91* (1–2), 21–31.
- (9) Stramare, S.; Thangadurai, V.; Weppner, W. Lithium Lanthanum Titanates: A Review. *Chem. Mater.* **2003**, *15* (21), 3974–3990.
- (10) Lutz, H. D.; Kuske, P.; Wussow, K. Ionic Motion of Tetrahedrally and Octahedrally Coordinated Lithium Ions in Ternary and Quaternary Halides. *Solid State Ionics* **1988**, *28–30*, Part 2, 1282–1286.
- (11) Cao, C.; Li, Z.-B.; Wang, X.-L.; Zhao, X.-B.; Han, W.-Q. Recent Advances in Inorganic Solid Electrolytes for Lithium Batteries. *Front. Energy Res.* **2014**, *2*, 1–10.
- (12) Takada, K. Progress and Prospective of Solid-State Lithium Batteries. *Acta Mat.* **2013**, *61* (3), 759–770.
- (13) Knauth, P. Inorganic Solid Li Ion Conductors: An Overview. *Solid State Ionics* **2009**, *180* (14), 911–916.
- (14) Kim, J. G.; Son, B.; Mukherjee, S.; Schuppert, N.; Bates, A.; Kwon, O.; Choi, M. J.; Chung, H. Y.; Park, S. A Review of Lithium and Non-Lithium Based Solid State Batteries. *J. Power Sources* **2015**, *282*, 299–322.
- (15) Yao, X.; Huang, B.; Yin, J.; Peng, G.; Huang, Z.; Gao, C.; Liu, D.; Xu, X. All-Solid-State Lithium Batteries with Inorganic Solid Electrolytes: Review of Fundamental Science. *Chinese Phys. B* **2016**, *25* (1), 018802.
- (16) Kamphorst, J. G.; Hellstrom, E. E. Fast Li Ionic Conduction in Solid Solutions of the System $\text{Li}_4\text{GeO}_4\text{-Li}_2\text{ZnGeO}_4\text{-Li}_3\text{PO}_4$. *Solid State Ionics* **1980**, *1* (3), 187–197.
- (17) Kuwano, J.; West, A. R. New Li^+ Ion Conductors in the System, $\text{Li}_4\text{GeO}_4\text{-Li}_3\text{VO}_4$. *Mater. Res. Bull.* **1980**, *15* (11), 1661–1667.
- (18) Wang, Y.; Richards, W. D.; Ong, S. P.; Miara, L. J.; Kim, J. C.; Mo, Y.; Ceder, G. Design Principles for Solid-State Lithium Superionic Conductors. *Nat Mater* **2015**, *14* (10), 1026–1031.
- (19) Ong, S. P.; Mo, Y.; Richards, W. D.; Miara, L.; Lee, H. S.; Ceder, G. Phase Stability, Electrochemical Stability and Ionic Conductivity of the $\text{Li}_{10\pm1}\text{MP}_2\text{X}_{12}$ ($\text{M} = \text{Ge, Si, Sn, Al}$ or P , and $\text{X} = \text{O, S}$ or Se) Family of Superionic Conductors. *Energy Environ. Sci.* **2013**, *6* (1), 148–156.
- (20) Kanno, R.; Murayama, M. Lithium Ionic Conductor Thio-LISICON: The $\text{Li}_2\text{S} - \text{GeS}_2 - \text{P}_2\text{S}_5$ System. *J. Electrochem. Soc.* **2001**, *148* (7), A742–A746.
- (21) Kraft, M. A.; Culver, S. P.; Calderon, M.; Böcher, F.; Krauskopf, T.; Senyshyn, A.; Dietrich, C.; Zevalkink, A.; Janek, J.; Zeier, W. G. Influence of Lattice Polarizability on the Ionic Conductivity in the Lithium Superionic Argpyrodites $\text{Li}_6\text{PS}_5\text{X}$ ($\text{X} = \text{Cl, Br, I}$). *J. Am. Chem. Soc.* **2017**, *139* (31), 10909–10918.
- (22) Almond, D. P.; West, A. R. Entropy Effects in Ionic Conductivity. *Solid State Ionics* **1986**, *18*, 1105–1109.

- (23) Yelon, A.; Movaghar, B.; Branz, H. M. Origin and Consequences of the Compensation (Meyer-Neldel) Law. *Phys. Rev. B* **1992**, *46* (19), 12244–12250.
- (24) Nowick, A. S.; Lee, W.-K.; Jain, H. Survey and Interpretation of Pre-Exponentials of Conductivity. *Solid State Ionics* **1988**, *28–30* (Part 1), 89–94.
- (25) Krauskopf, T.; Pompe, C.; Kraft, M. A.; Zeier, W. G. Influence of Lattice Dynamics on Na⁺ Transport in the Solid Electrolyte Na₃PS_{4-x}Se_x. *Chem. Mater.* **2017**, *29* (20), 8859–8869.
- (26) Almond, D. P.; West, A. R. The Activation Entropy for Transport in Ionic Conductors. *Solid State Ionics* **1987**, *23* (1–2), 27–35.
- (27) Ngai, K. L. Meyer–Neldel Rule and Anti Meyer–Neldel Rule of Ionic Conductivity: Conclusions from the Coupling Model. *Solid State Ionics* **1998**, *105* (1), 231–235.
- (28) Yelon, A.; Movaghar, B.; Crandall, R. S. Multi-Excitation Entropy: Its Role in Thermodynamics and Kinetics. *Rep. Prog. Phys.* **2006**, *69* (4), 1145–1194.
- (29) Dienes, G. J. Frequency Factor and Activation Energy for the Volume Diffusion of Metals. *J. Appl. Phys.* **1950**, *21* (11), 1189–1192.
- (30) Deng, Y.; Eames, C.; Fleutot, B.; David, R.; Chotard, J.-N.; Suard, E.; Masquelier, C.; Islam, M. S. Enhancing the Lithium Ion Conductivity in Lithium Superionic Conductor (LISICON) Solid Electrolytes through a Mixed Polyanion Effect. *ACS Appl. Mater. Interfaces* **2017**, *9* (8), 7050–7058.
- (31) Tachez, M.; Malugani, J.-P.; Mercier, R.; Robert, G. Ionic Conductivity of and Phase Transition in Lithium Thiophosphate Li₃PS₄. *Solid State Ionics* **1984**, *14* (3), 181–185.
- (32) Hu, Y.-W.; Raistrick, I. D.; Huggins, R. A. Ionic Conductivity of Lithium Orthosilicate–Lithium Phosphate Solid Solutions. *J. Electrochem. Soc.* **1977**, *124* (8), 1240–1242.
- (33) Shannon, R. D. Revised Effective Ionic Radii and Systematic Studies of Interatomic Distances in Halides and Chalcogenides. *Acta Crystallogr. A* **1976**, *32* (5), 751–767.
- (34) Plimpton, S. Fast Parallel Algorithms for Short-Range Molecular Dynamics. *J. Comput. Phys.* **1995**, *117* (1), 1–19.
- (35) Pedone, A.; Malavasi, G.; Menziani, M. C.; Cormack, A. N.; Segre, U. A New Self-Consistent Empirical Interatomic Potential Model for Oxides, Silicates, and Silica-Based Glasses. *J. Phys. Chem. B* **2006**, *110* (24), 11780–11795.
- (36) Broglia, G.; Mugoni, C.; Du, J.; Siligardi, C.; Montorsi, M. Lithium Vanado-Phosphate Glasses: Structure and Dynamics Properties Studied by Molecular Dynamics Simulations. *J. Non-Cryst. Solids* **2014**, *403*, 53–61.
- (37) Deng, Y.; Eames, C.; Chotard, J.-N.; Lalère, F.; Seznec, V.; Emge, S.; Pecher, O.; Grey, C. P.; Masquelier, C.; Islam, M. S. Structural and Mechanistic Insights into Fast Lithium-Ion Conduction in Li₄SiO₄–Li₃PO₄ Solid Electrolytes. *J. Am. Chem. Soc.* **2015**, *137* (28), 9136–9145.
- (38) Ong, S. P.; Richards, W. D.; Jain, A.; Hautier, G.; Kocher, M.; Cholia, S.; Gunter, D.; Chevrier, V. L.; Persson, K. A.; Ceder, G. Python Materials Genomics (Pymatgen): A Robust, Open-Source Python Library for Materials Analysis. *Comput. Mater. Sci.* **2013**, *68*, 314–319.
- (39) Deng, Z.; Zhu, Z.; Chu, I.-H.; Ong, S. P. Data-Driven First-Principles Methods for the Study and Design of Alkali Superionic Conductors. *Chem. Mater.* **2017**, *29* (1), 281–288.
- (40) Hodge, I. M.; Ingram, M. D.; West, A. R. Ionic Conductivity of Li₄SiO₄, Li₄GeO₄, and Their Solid Solutions. *J. Am. Ceram. Soc.* **1976**, *59* (7–8), 360–366.
- (41) Ivanov-Shitz, A. K.; Kireev, V. V.; Mel'nikov, O. K.; Demianets, L. N. Growth and Ionic Conductivity of γ- Li₃PO₄. *Crystallogr. Rep.* **2001**, *46* (5), 864–867.
- (42) Bruce, P. G.; West, A. R. The A-C Conductivity of Polycrystalline LISICON, Li_{2+2x}Zn_{1-x}GeO₄, and a Model for Intergranular Constriction Resistances. *J. Electrochem. Soc.* **1983**, *130* (3), 662–669.
- (43) Krauskopf, T.; Culver, S. P.; Zeier, W. G. Bottleneck of Diffusion and Inductive Effects in Li₁₀Ge_{1-x}Sn_xP₂S₁₂. *Chem. Mater.* **2018**, *30* (5), 1791–1798.
- (44) Balluffi, R. W. *Kinetics of Materials*; Wiley-Interscience: Hoboken, N.J., 2005.
- (45) Muy, S.; Bachman, J. C.; Giordano, L.; Chang, H.-H.; Abernathy, D. L.; Bansal, D.; Delaire, O.; Hori, S.; Kanno, R.; Maglia, F.; et al. Tuning Mobility and Stability of Lithium Ion Conductors Based on Lattice Dynamics. *Energy Environ. Sci.* **2018**, *11*, 850–859.
- (46) Isayev, O.; Oses, C.; Toher, C.; Gossett, E.; Curtarolo, S.; Tropsha, A. Universal Fragment Descriptors for Predicting Properties of Inorganic Crystals. *Nat. Commun.* **2017**, *8*, 15679.
- (47) Du, Y.; Holzwarth, N. Mechanisms of Li⁺ Diffusion in Crystalline γ- and β- Li₃PO₄ Electrolytes from First Principles. *Phys. Rev. B* **2007**, *76* (17), 174302–174315.
- (48) Bruce, P. G.; West, A. R. Ionic Conductivity of LISICON Solid Solutions, Li_{2+2x}Zn_{1-x}GeO₄. *J. Solid State Chem.* **1982**, *44* (3), 354–365.
- (49) Winand, J.-M.; Rulmont, A.; Tarte, P. Nouvelles Solutions Solides L⁺(M^{IV})_{2-x}(N^{IV})_x(PO₄)₃ (L = Li, Na; M, N = Ge, Sn, Ti, Zr, Hf) Synthèse et Étude Par Diffraction x et Conductivité Ionique. *J. Solid State Chem.* **1991**, *93* (2), 341–349.
- (50) Sahu, G.; Rangasamy, E.; Li, J.; Chen, Y.; An, K.; Dudney, N.; Liang, C. A High-Conduction Ge Substituted Li₃AsS₄ Solid Electrolyte with Exceptional Low Activation Energy. *J. Mater. Chem. A* **2014**, *2* (27), 10396–10403.
- (51) Sahu, G.; Lin, Z.; Li, J.; Liu, Z.; Dudney, N.; Liang, C. Air-Stable, High-Conduction Solid Electrolytes of Arsenic-Substituted Li₄SnS₄. *Energy Environ. Sci.* **2014**, *7* (3), 1053–1058.
- (52) Kato, Y.; Saito, R.; Sakano, M.; Mitsui, A.; Hirayama, M.; Kanno, R. Synthesis, Structure and Lithium Ionic Conductivity of Solid Solutions of Li₁₀(Ge_{1-x}M_x)P₂S₁₂ (M = Si, Sn). *J. Power Sources* **2014**, *271*, 60–64.
- (53) Kamaya, N.; Homma, K.; Yamakawa, Y.; Hirayama, M.; Kanno, R.; Yonemura, M.; Kamiyama, T.; Kato, Y.; Hama, S.; Kawamoto, K.; et al. A Lithium Superionic Conductor. *Nat. Mater.* **2011**, *10* (9), 682–686.
- (54) Cussen, E. J.; Yip, T. W. S.; O'Neill, G.; O'Callaghan, M. P. A Comparison of the Transport Properties of Lithium-Stuffed Garnets and the Conventional Phases Li₃Ln₃Te₂O₁₂. *J. Solid State Chem.* **2011**, *184* (2), 470–475.
- (55) Narayanan, S.; Thangadurai, V. Effect of Y Substitution for Nb in Li₃La₃Nb₂O₁₂ on Li Ion Conductivity of Garnet-Type Solid Electrolytes. *J. Power Sources* **2011**, *196* (19), 8085–8090.
- (56) Murugan, R.; Thangadurai, V.; Weppner, W. Lattice Parameter and Sintering Temperature Dependence of Bulk and Grain-Boundary Conduction of Garnet-like Solid Li-Electrolytes. *J. Electrochem. Soc.* **2008**, *155* (1), A90–A101.
- (57) Thangadurai, V.; Weppner, W. Li₆Ala₂Nb₂O₁₂ (A=Ca, Sr, Ba): A New Class of Fast Lithium Ion Conductors with Garnet-Like Structure. *J. Am. Ceram. Soc.* **2005**, *88* (2), 411–418.
- (58) Zhong, Y.; Zhou, Q.; Guo, Y.; Li, Z.; Qiang, Y. The Ionic Conductivity of Li₆BaLa₂M₂O₁₂ with Coexisting Nb and Ta on the M Sites. *Ionics* **2013**, *19* (4), 697–700.

